

bars, to those meant for biomedical applications, which can measure a few millibars of pressure.

There is also a special biomedical pressure sensor that they have developed, which is implanted into the skull during neurosurgery. It continuously measures intracranial pressure, and is extremely useful when doctors are performing neurosurgery. This technology is readily available for people to commercialise.

Dr Mishra also talked about a solid-state gas sensor platform. The gas sensors contain a micro-heater that consumes only 13 to 14 milliwatts of energy. It is a power-efficient device and has a quick response time.

An electrode is deposited over the micro-heater. Over the electrode there is a sensing element that has a nano-structured film whose resistance changes when it is exposed to a particular gas. The film is made using carbon nano-particles, and is the key to the kind of sensitivity the device can have. Change in resistance is a function of concentration of gas.

Finally, a substrate layer is etched from the bottom. Thickness of the film is 13 to 14 microns. This is required so that the micro-heater does not take a long time to heat up. If the substrate layer is thick, a lot of heat gets dissipated, and the sensor uses more power. Once the device is ready, it is taken into a known ambient of that specific gas, and the sensor is tested.

High-tech health

An interesting device that has been fabricated at CeNSE lab and now has one United States patent and seven Indian patents is a diabetes sensor. Dr Navakanta Bhat has developed it to aid in diabetic management as well as to understand the long-term effects of diabetes on other organs of the body.

Ophthalmic OCT on a Chip (OCTCHIP) is a project started in 2016 that involves optical coherence tomography (OCT) systems to be built much cheaper through the use of photonic integrated circuits. The solution has

20 exciting sensor applications for a smarter world

Structural health. Monitoring vibrations and material conditions in bridges and buildings

Noise urban maps. Real-time sound monitoring in cities and zones

Traffic congestion. Monitoring vehicles and pedestrians to optimise driving and walking routes

Smartlighting. Intelligent and weather adaptive lighting in streetlights

Waste management. Detection of rubbish levels in containers to optimise trash-collection routes

Smartroads. Intelligent highways with warning messages and diversions according to climate and unexpected events

Earthquake early detection. Distributed control in specific places of tremors

Water leakages. Detection of liquid presence outside tanks and pressure variations along pipes

Photovoltaic installations. Monitoring and optimisation of performance in solar energy plants

Perimeter access control. Access control to restricted areas and detection of people in non-authorised areas

Smartproduct management. Control of rotation of products in shelves and warehouses to automate restocking processes

Quality of shipment conditions. Monitoring of vibrations, strokes, container openings or cold chain maintenance for insurance purposes

Vehicle auto-diagnosis. Information collection from CanBus to send real-time alarms for emergencies or provide advice to drivers

Greenhouses. Control micro-climate conditions to maximise production of fruits and vegetables and their quality

Hydroponics. Control the exact conditions of plants grown in water to get the highest efficiency crops

Sportsmen care. Vital signs monitoring in high-performance centres and fields

Patients surveillance. Monitoring of conditions of patients inside hospitals and in old people's homes

Ultraviolet. Measurement of ultraviolet sun rays to warn people not to be exposed in certain hours

Radiation levels. Distributed measurement of radiation levels in nuclear power stations surroundings to generate leakage alerts

Air pollution. Control of carbon-dioxide emissions of factories, pollution emitted by cars and toxic gases generated in farms

a swept source, electrical interposer, photonic integrated circuit, micro-lens array and reference arm on a chip smaller than the new five-rupee coin. This project is funded by European Union's Horizon 2020 research and innovation programme.

Wireless sensor networks

Wireless sensor nodes can be used to improve functionality and convenience of applications. Without wires to hinder installation, these sensors can be placed at strategic locations to continuously collect data on their surroundings. This makes these a perfect tool for building systems in line with the IoT concept where systems can autonomously collect data, process and actuate it in response to the information gleaned out of the data.

Fraunhofer has a technology called s-net that lets engineers create energy-saving wireless networks

for long-lasting large-scale systems and smartobjects in an IoT network. Inside s-net, proprietary media-access protocol SlottedMAC combines the efficiency advantages of a time division multiple access system with scalability and flexibility of contention based protocols to enable an intermeshed networking of sensor nodes.

"Intermeshed networking creates the basis for accurate localisation of sensor nodes. Based on an enhanced weighted centroid localisation algorithm, the sensor node independently calculates its own position from the positions of all received nodes. As a result, the position is calculated in such a way that resources are spared, and this can be signalled to a central entity upon request or at set time periods. As the position is calculated directly on the node, applications can also trigger an alarm without network communication, for example, in